

DOAN GIA BAO

UNITY GAME DEVELOPER

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[portfolio](#) | [github](#)

Dedicated Software Engineer with a rigorous focus on game architecture and performance optimization, committed to building high-efficiency systems that power seamless gameplay experiences.

Looking ahead, I am committed to evolving into a Technical Artist, specializing in Shader programming and VFX to bridge the gap between technical constraints and high-fidelity visual excellence.

PROFESSIONAL EXPERIENCE

Thetan Immortal - PvP Archer

Aug 2025 - Nov 2025

Wolffun Game Studio - Junior Unity Game Developer

Collaborating with mentors and team leaders within a cross-functional team - Design, Art, Data.

- Using OSA (Optimized Scroll-View Adapter) for Shop and Leaderboard systems to eliminate frame drops and memory spikes when rendering large-scale dynamic lists.
- Utilized ScriptableObjects for Data-driven Event systems to decouple UI logic from gameplay, ensuring a highly modular and easily maintainable codebase.
- Using Unity Localization Package with Google Sheet API for all UI and in-game dialogue, streamlining the translation workflow and reducing manual content updates.
- Integrated a QR code referral system via API to enable user acquisition tracking and seamless in-game referral functionality.
- Engineered a modular Skill System using Unity Behavior Tree architecture to decouple combat logic from AI decision-making, allowing for rapid iteration and scalable bot abilities.

Project winner of [Google Play's Best of 2025](#) award, demonstrating excellence in game design, technical execution, and product representing the best of homegrown talent.

Tile Travel - Tile Triple Match

Nov 2025 - Present

Wolffun Game Studio - Junior Unity Game Developer

Solo Developer within a cross-functional team - Design, Art, Data.

- Using Object Pooling and Pattern-matching algorithms for tile spawning and match-3 mechanics, ensuring zero-lag gameplay and efficient memory management during high-speed tile clearing.
- Architected a ScriptableObject-based framework for modular game behaviors, enabling rapid mechanics integration through decoupled, data-driven logic.
- Developed a Tile-Distribution Algorithm for the Level Journey system, automating icon placement to accelerate level design workflows while ensuring strictly controlled difficulty curves.
- Developed Odin-based/Custom Property Drawers to manage Complex Gameplay Configurations, ensuring a structured and error-proof data entry workflow for non-technical designers.
- Optimized CPU performance by implementing Culling techniques and DOTween Animation, reducing calculation overhead and providing smooth visual feedback during high-intensity gameplay.
- Implemented innovative gameplay mechanics and dynamic event-driven game modes to enhance player engagement, retention, and long-term replay value.
- Leveraged Ad Mediation frameworks (AppLovin/AdMob) to integrate Rewarded and Interstitial ads, optimizing ad placement frequency to maximize eCPM without compromising player retention.
- Implemented Custom Event Tracking using Firebase Analytics to monitor player funnels and level drop-off rates, providing actionable insights to optimize game difficulty and player progression.

Project recognized as the top-performing product in key metrics, delivering strong retention, solid monetization, and proven scalability for large-scale growth.

EDUCATION

Bachelor of Software Engineering | Frontend Development Specialist

Aug 2020 - Oct 2024

FPT University

- Strong foundation in data Structures & Algorithms, Object-Oriented Programming (OOP).
- Expertise in Component-based Architecture.
- Skilled in architecting real-time systems using RESTful APIs and Web Sockets for seamless client-server data synchronization.

Developed E-Furniture, an AR app for real-time 3D furniture visualization in home environments.

Academy-Trained Game Developer | Unity 2D/3D Specialist

Oct 2022 - Aug 2025

VTC Academy

- Experienced in implementing Rigidbodies, Colliders, and Raycasting for precise object interaction.
- Strong grasp of Linear Algebra (Vectors, Quaternions, Dot/Cross Products) for spatial positioning, rotation, and movement logic.
- Leveraging ScriptableObjects to architect data-driven systems, decoupling game logic from data.

Developed a 2D Triple Tile Match puzzle game, implementing optimized grid-based algorithms.

Developed a 3D FPS Zombie Shooter with a focus on immersive shooting mechanics, leveraging Raycasting for precise hit detection.

PERSONAL DEVELOPMENT & TECHNICAL EXPLORATION

System Design

- Designing scalable game architecture using SOLID principles, and modular feature-based structure.
- Applying Dependency Injection, Observer Pattern, and Event Bus patterns to build loosely coupled and testable systems.

Game Optimization

- Advanced profiling with Unity Profiler, Frame Debugger, and Memory Profiler to identify CPU spikes, GC allocations, and overdraw.
- Implementing Object Pooling, Flyweight Pattern, and memory management strategies to minimize runtime allocations.
- Optimizing rendering via batching strategies, draw call reduction, and GPU Instancing within mobile performance constraints.

Shader & VFX

- Developing custom shaders using Shader Graph and HLSL for procedural effects (distortion, blur, swirl, UV manipulation).
- Understanding GPU pipeline stages, render queues, and material instancing optimization.
- Implementing performance-aware VFX using particle systems and shader-based animation techniques.

AI-Assisted Engineering Workflow

- Integrating Claude into architecture reviews, refactoring strategies, and algorithm optimization.
- Designing structured prompt frameworks and reusable Claude Skill workflows for engineering automation.
- Applying MCP-based workflow orchestration to streamline debugging, documentation, and system iteration.

ADDITIONAL INFORMATION

- **Languages:** English (Academic Preparatory Certification - FPT University).
- **Achievements:** Top 5 Best Projects - Rocket Studio Game Jam.
- **Interests:** Graphics Optimization, Shader HLSL, Procedural Level Generation.